Arista Access Node BAS-ACC-54 Hardware/Environmental Fault Troubleshooting and Resolution SOP

Node: BAS-ACC-54 | **Vendor:** Arista | **Severity:** Critical | **Alarm Code:** ALARM_6085

Description

The Arista Access Node BAS-ACC-54, deployed at the Basavanagudi POP, is an Arista switch operating in the Access role. It utilizes an Arista EOS (Extensible Operating System) to manage network traffic. The failure domain in question revolves around Hardware/Environmental Faults, specifically ASIC memory parity errors, line card crashes, and power supply redundancy loss. This failure domain occurs due to the inherent complexity of the Access Node's design, which involves the convergence of multiple protocol stacks, including TCP/IP, BGP, OSPF, and EIGRP, on a single ASIC. The ASIC memory parity errors and line card crashes are symptoms of hardware malfunctions, while power supply redundancy loss is a result of environmental factors such as power outages or equipment failure. The Arista EOS, in an effort to mitigate these failures, triggers the ALARM_6085 alarm, signaling a critical failure condition.

Pre-requisite Checks

- Verify that all redundant links are functioning correctly.

- Check the power supply redundancy status to ensure at least one power supply is operational.

- Dispatch remote hands to the Basavanagudi POP to physically check the Node for any visible signs of damage or environmental stress.

- Gather all relevant logs and event records from the Node and surrounding equipment.

Potential Root Causes

Cause	Probability
ASIC memory parity error	High
Line card crash	Medium
Power supply redundancy loss	Low

Diagnostic Commands & Expected Outputs

```
Command: show system internal switch internals, Expected Output: Switch internals information, including ASIC memory usage, line card status, and power supply redundancy status., Description: This command is used to gather information about the internal switch state, including ASIC memory usage, line card status, and power supply redundancy status. It helps troubleshoot issues related to ASIC memory parity errors, line card crashes, and power supply redundancy loss.
```

```
Command: show logging facility 4, Expected Output: System log messages related to critical failures, including ASIC memory parity errors, line card crashes, and power supply redundancy loss., Description: This command is used to retrieve system log messages related to critical failures, including ASIC memory parity errors, line card crashes, and power supply redundancy loss. It helps identify the root cause of the failure.
```

Resolution Steps

Step 1: Step: Replace the faulty ASIC, Command: system internal switch replace asic, Description: Replace the faulty ASIC with a new one, ensuring that the replacement process is performed by a qualified technician.

Step 2: Step: Replace the faulty line card, Command: system internal switch replace linecard, Description: Replace the faulty line card with a new one, ensuring that the replacement process is performed by a qualified technician.

Step 3: Step: Replace the power supply, Command: system internal switch replace power-supply, Description: Replace the faulty power supply with a new one, ensuring that the replacement process is performed by a qualified technician.

Post-Check Verification

```
Command: show system internal switch internals, Expected Output: Switch internals information, including ASIC memory usage, line card status, and power supply redundancy status., Description: Verify that the switch internals information shows no signs of hardware or environmental faults.

Command: show logging facility 4, Expected Output: System log messages indicating normal system operation., Description: Verify that the system log messages show no signs of critical failures.
```

Roll-back Procedures

- Step: Restore the original ASIC, Command: system internal switch restore asic, Description: Restore the original ASIC, ensuring that the replacement process is performed by a qualified technician.

- Step: Restore the original line card, Command: system internal switch restore linecard, Description: Restore the original line card, ensuring that the replacement process is performed by a qualified technician.

- Step: Restore the original power supply, Command: system internal switch restore power-supply, Description: Restore the original power supply, ensuring that the replacement process is performed by a qualified technician.